
Sort-and-Count(L)

 If the list has one element then
 there are no inversions

Else

 Divide the list into two halves:

A contains the first $\lfloor n/2 \rfloor$ elements

B contains the remaining $\lfloor n/2 \rfloor$ elements

$(r_A, A) = \text{Sort-and-Count}(A)$

$(r_B, B) = \text{Sort-and-Count}(B)$

$(r, L) = \text{Merge-and-Count}(A, B)$

Endif

Return $r = r_A + r_B + r$, and the sorted list L
